SPECTRA: A Traceability Ontology for the 3GPP RAN Standardization Process

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Abstract. Standardization documents are produced through a dense, interlinked, and continuously revised workflow of technical contributions, working-group decisions, and change requests. Yet this standardization process is rarely captured as a queryable knowledge graph in the Semantic Web, leaving traceability tasks (recovering the contribution lineage behind a clause, mapping cross-group Liaison-Statement chains, aggregating per-company contribution patterns) to manual archive search. We present **SPECTRA** (*SPECification TRAcing*), an OWL ontology of the 3GPP Radio Access Network (RAN) standardization process. The schema is organized around four concerns common to standardization workflows — contributions, resolutions, specifications, and cross-group communication — and was developed iteratively through cumulative competency-question regression. We validated reuse by instantiating the RAN1-designed schema across all five RAN Working Groups, which required no additional classes. The release accompanies *SpectraCQ v1.0*, a 137-question competency-benchmark dataset citable as a standalone subresource. SPECTRA-authored components (ontology, SHACL shapes, documentation, queries, examples, validation scripts, and the sanitized parsing pipeline) are released under CC-BY 4.0; the bundled per-WG body-text knowledge graphs are 3GPP-derived literals redistributed with explicit 3GPP attribution.

Keywords: Ontology engineering · OWL · 3GPP · Standardization process · Document traceability · Competency questions · Semantic Web

1 Introduction

Standards-developing organizations (SDOs) such as the 3rd Generation Partnership Project (3GPP), the Internet Engineering Task Force (IETF), and the Institute of Electrical and Electronics Engineers (IEEE) produce evolving document corpora. The lifecycle of these documents—from initial contributions through working-group (WG) resolutions to normative specification text—shapes which technologies become global infrastructure. The Semantic Web has extensively modeled provenance [10] and related publication or software

artifacts, but the SDO standardization *process itself* remains under-modeled, even though SDOs produce artifacts that govern interoperability across the telecom, networking, and semiconductor industries. We focus on a large and practically important under-modeled SDO setting: 3GPP, the global SDO defining 4G Long-Term Evolution (LTE) and 5G NR (New Radio), and coordinating ongoing 6G standardization. The Radio Access Network (RAN) Technical Specification Group (TSG) alone encompasses five Working Groups (RAN1–RAN5), each responsible for a distinct layer of the radio interface, from physical-layer procedures to conformance testing.

3GPP specifications are the normative reference that commercial 5G implementations and conformance tests must align with. Yet the corpus spans hundreds of normative specifications and evolves through a process that produces hundreds of thousands of Working Group documents. Acquiring even partial working knowledge typically requires substantial domain expertise and prolonged immersion. This creates barriers along three axes: *cross-WG*, where a RAN1 decision can propagate to the sibling RAN WGs (RAN2–RAN5) through Liaison Statements (LSs) that are difficult to track end-to-end; *cross-organization*, where each company has internal context mainly for its own contributions; and *industry–academia*, where researchers see the published specifications but not the meeting-level decisions that shaped them. The traceability path TDoc $\rightarrow$ Resolution $\rightarrow$ CR $\rightarrow$ TS Section $\rightarrow$ Spec, which helps recover the document-level lineage behind a clause and locate where it resides, currently exists as implicit institutional knowledge and is typically reconstructed by hand — error-prone and time-consuming.

To our knowledge in the venues we surveyed (ISWC, ESWC, JWS, IEEE Globecom 2020–2025), no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle. Recent 3GPP-domain resources — TSpec-LLM [11] for retrieval-augmented access to specification text and *telecom-kg-rel19* [23], a GraphML entity–relation KG with provenance and text chunks built from Release-19 specification text — model the *content* of specifications but are not designed to directly represent the standardization *process* that produced them. Examples of such process-level questions include: which TDocs led to a clause, which CRs responded to or were linked from a Liaison Statement, or which Working Assumptions were promoted to Agreements.

In this paper, we present **SPECTRA** (*SPECification TRAcing*), an OWL resource for modeling the types, relationships, and lifecycle of 3GPP RAN standardization documents. The schema's four-concern structure (contributions, resolutions, specifications, and cross-group communication) appears to map onto sibling SDOs whose document lifecycles share the same shape (contributions are tabled, working-group resolutions decide their fate, change requests amend the normative text, and cross-group communication propagates impact); empirical validation outside 3GPP RAN is left to future work. The contributions are:

1. **Traceability tasks SPECTRA enables** — a small but practically important family of cross-document queries that are otherwise reconstructed by hand: recovering the document-level lineage of a clause's current text

(S1, multi-hop trace), mapping cross-WG Liaison-Statement chains (S2), aggregating Release-scoped CR analytics (S3), surfacing Working-Assumption promotion lineage (S4), propagating TR-to-TS impact (S5), per-company contribution profiles (S6), and Work-Item-scoped lineage (S7). Each task is shown in §6 with the schema fragment it exercises.

2. **Public release package** — SPECTRA-authored components (the OWL ontology with 26 classes, 53 object properties, 81 data properties; reuses Dublin Core, DCTERMS, FOAF; optional PROV-O alignment; SHACL shapes, PyLODE documentation, representative Cypher/SPARQL queries, synthetic and metadata-only examples, a cross-WG metadata-only process KG over RAN1–RAN5, and the sanitized parsing pipeline source) under CC-BY 4.0, alongside per-WG body-text KGs as 3GPP-derived literals retained with explicit 3GPP attribution following [11,23]; release details and licensing tiers in Section 7.
3. ***SpectraCQ v1.0* as a standalone sub-resource** — the 137-CQ dataset (natural-language (NL) question + executable Cypher + RAN1 verdict; phase distribution 25/34/45/15/18) is released as a separately citable CC-BY 4.0 artifact intended for reuse as a CQ-to-query test-suite for SDO-process ontologies independent of SPECTRA.
4. **Cross-WG and reproducibility-aware validation evidence** — the schema is instantiated at realistic scale across all five RAN Working Groups; applying the RAN1-designed schema to RAN2–RAN5 required *no additional classes beyond the RAN1-designed schema*, only three schema-level extension candidates (one object property and two data properties: `summaryId` from RAN3, `discussionText` from RAN5; Table 7). The release ships the schema, validation JSONs, the cross-WG process KG, per-WG body-text KGs (RAN1–RAN5, with 3GPP attribution), and the sanitized parsing pipeline; reviewers can re-derive the artifact-integrity claims directly via `verify_release.py`, while full 137-CQ Cypher correctness depends on the operational KG and is recorded in `validation/cq_results.json`. Raw Neo4j dumps and VectorDB embeddings are excluded (large binary artifacts; regenerable from the bundled artifacts).
5. **Cumulative CQ quality-assurance practice** — given the scale of 3GPP RAN (hundreds of specifications, hundreds of thousands of TDocs across five WGs), we developed SPECTRA through five incremental phases (P1–P5; 25/34/45/15/18 CQs, 137 total; phase-to-document mapping in §5) with cumulative regression checks at each transition; three QA cases surfaced and were resolved before release. This practice is complementary to established CQ-driven methodologies (LOT [16], SAMOD [13], NeOn [20]) rather than a replacement.

2 Related Work

2.1 Ontologies for Domain Knowledge Management

Ontologies have been widely adopted as the semantic backbone for domain-specific knowledge management systems. In cultural heritage, the Polifonia On-

tology Network [2] provides 15 modular ontologies for musical heritage, developed through eXtreme Design [18] with 361 competency questions from 10 pilots. CQ-driven design also appears in the Hydrogen Ontology (HOLY) [1] for energy markets. In industry, Cao et al. [4] describe a knowledge graph for the quality management of Bosch products as an example of industrial KG deployment. For technical standards, Grangel-González and Vidal [6] use the Standards Ontology (STO) as the schema for an Industry 4.0 Knowledge Graph covering more than 200 industrial standards, but their focus is on *inter-standard* relationships rather than the *standardization-process document lifecycle* (sections, change requests, meeting decisions, liaison statements) that is central to our work. In telecommunications, NORIA-O [21] targets anomaly detection and incident management in ICT systems rather than the standardization process itself. Closer in spirit, formal document lifecycles have been modelled outside telecommunications by UNDO [14] (UN documents, generic Workflow vocabulary) and the IETF RFC ontology [15] (published-RFC bibliographic metadata: `obsoletes`/`updates`/`Errata`); neither targets an SDO's working-group deliberation events (meeting-level resolutions, cross-WG liaison routing, working-assumption-to-agreement promotions), which are the SPECTRA core. A recent effort in the 3GPP domain is *telecom-kg-rel19* [23], a Hugging Face dataset distributing a GraphML entity–relation KG with provenance and text chunks derived from Release-19 specification *text* for retrieval-augmented LLM reasoning. Unlike that resource, SPECTRA is an OWL ontology of the standardization process and document lifecycle: it models TDoc, Resolution, CR, LS, TR, Section, and related process entities rather than concept structures embedded in specification text. In the venues we surveyed (ISWC, ESWC, JWS, IEEE Globecom 2020–2025), no publicly available OWL ontology specifically targets the 3GPP RAN standardization document lifecycle and traceability tasks. Table 1 positions SPECTRA against the closest comparable resources along the axes most relevant to the ISWC Resource Track.

2.2 Ontology Engineering Methodologies

Ontology 101 [12] popularized a practical ontology-development workflow, and competency questions (CQs) have long served as the primary requirements mechanism [7]. LOT [16] extends this for industrial settings with FAIR compliance. The NeOn methodology [20] provides scenario-based pathways for building ontology networks, including reuse of existing ontological and non-ontological resources. SAMOD [13] draws on Test-Driven Development principles. SPECTRA's five-phase protocol (§5) is complementary rather than a replacement: the accumulated CQ suite is re-executed at each internal phase transition so earlier-phase competencies are revisited each time the schema grows. This practice surfaced three regression cases (§5) that single-phase validation would not have detected before release.

In terms of CQ scale, the Polifonia Ontology Network [2] releases a 361-CQ dataset (*PolifoniaCQ*) collected across the network's modules, and Wisniewski et al. [22] translated 234 CQs into 131 SPARQL-OWL queries. SPECTRA contributes 137 CQs applied cumulatively to a single cohesive ontology spanning five

Table 1. Comparison with related Semantic Web resources along Resource-Track-relevant axes (each work's published abstract / availability statement). *Process* = explicit domain process / lifecycle model; *multi-hop trace* = backward traceability over $\geq$ 3 class types; *CQ-grounded* = explicit CQ set; *cross-WG/subdomain* = released schema validated across $\geq$ 2 sibling WGs or subdomains in the stated domain. For SPECTRA this is RAN1–RAN5 of the RAN TSG (one schema, five sibling WGs); for Polifonia, ontology-network modules covering distinct musical subdomains.

Resource	Domain	Process	Multi-hop trace	CQ-grounded	Cross-WG/ subdomain
Polifonia [2]	cultural heritage	limited	n/a	yes (361 CQs)	yes (15 modules)
HOLY [1]	energy markets	limited	n/a	yes (CQ tests)	n/a
Bosch QM [4]	industrial QM	limited	n/a	not reported	n/a
NORIA-O [21]	ICT incidents	yes (incident)	yes (incident RCA)	yes (NOC/SOC)	n/a
Grangel-G. & Vidal [6]	Industry 4.0 stds	limited	n/a	not reported	n/a
UNDO [14]	UN documents	yes (workflow)	limited	not reported	n/a
RFC ontology [15]	RFC metadata	limited	limited (updates)	not reported	n/a
TSpec-LLM [11]	3GPP text	n/a	n/a	n/a	n/a
telecom-kg-rel19 [23]	3GPP Rel-19 text	n/a	n/a	n/a	n/a
SPECTRA (ours)	**3GPP RAN proc.**	**yes**	**yes (5-class, join-aware)**	**yes (137 CQs)**	**yes (5 WGs)**

Axes chosen to surface differences in process modelling, multi-hop traceability, and CQ-grounded design relevant to the Resource Track rubric. "n/a" means the axis is outside the cited resource's stated design focus, not a claim that the resource is incapable of it. Under these axes, SPECTRA is the only listed resource focused on 3GPP RAN process modelling with document-lifecycle traceability and cross-WG instantiation evidence.

document-lifecycle areas: TDoc metadata, meeting resolutions, TS structure, CR documents, and technical reports.

3 3GPP Standardization Background

3.1 Document Types and Lifecycle

SPECTRA models the main interlinked artifact families in the 3GPP RAN document lifecycle (Figure 1): TDocs (with CR and LS subtypes), Resolutions, Specifications/Sections, and Technical Reports. **TDoc** is the primary contribution unit submitted as a temporary document to a meeting (e.g., R1-2400001); **CR** (Change Request modifying normative text) and **LS** (Liaison Statement carrying communication to other 3GPP groups or to external technical bodies) are subtypes of TDoc per 3GPP's own usage. **Resolution** captures meeting decisions (Agreement/Conclusion/Working Assumption) and is modelled as a first-class SPECTRA superclass so that per-meeting decisions are queryable regardless of subtype. **TS** (Technical Specification, e.g., TS 38.211) carries the normative text in hierarchical Sections; **TR** (Technical Report — informative study/feasibility material) links to the Specs it influences via reified `TRImpact`.

3.2 Working Group Structure

The RAN TSG is divided into five WGs whose responsibilities follow 3GPP's official scope statements: **RAN1** (Radio Layer 1, e.g., TS 38.211–214); **RAN2** (Radio Layer 2 protocols and Radio Resource Control, e.g., TS 38.331/321); **RAN3** (NG-RAN architecture and related interfaces, e.g., TS 38.401/423); **RAN4** (Radio Performance and Protocol Aspects, e.g., TS 38.101/104); **RAN5** (Mobile terminal conformance testing, e.g., TS 38.521/533). Inter-group dependencies are mediated by Liaison Statements addressed to other 3GPP WGs and

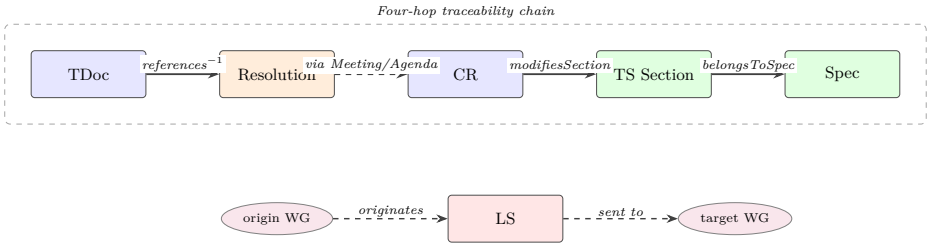

Fig. 1. Document lifecycle in 3GPP RAN. The top row is a reader-facing traceability path from a TDoc contribution to the affected specification: `references` is formally defined Resolution → Tdoc and is drawn in the reading direction, `modifiesSection` links CRs to Sections, and `belongsToSpec` links Sections to Specs. The dashed Resolution–CR step denotes an indirect join through meeting/agenda and CR-pack context rather than a single asserted ontology property. The bottom row summarizes LS-based cross-WG communication, also drawn in reader-facing direction; the formal properties are `originatedFrom` (LS → origin WG) and `sentTo` (LS → target WG). The sixth document type, *TR*, reaches the spine through the reified `TRImpact` relation shown in Figure 2.

to external technical bodies; RAN WG meetings convene on an approximately two-month cadence (varying 6 weeks to 3 months, with e-meetings during 2020–2022) and may involve thousands of TDocs.

Generalization beyond 3GPP (hypothetical). Table 2 sketches a *hypothetical analogy* to IETF and IEEE because the four-concern shape (contributions, resolutions, specifications, and cross-group communication) appears at face value to be common. The IETF/IEEE rows are loose document-class analogues, not property-level mappings; a real instantiation with its own elicited CQs is required to confirm reuse.

Table 2. Hypothetical four-concern analogy across three SDOs (not empirically validated). The lifecycle shape (contribution → resolution → amendment → specification/report artifact) appears comparable; vocabulary and process specifics differ, and the IETF/IEEE rows are illustrative document-class analogues only.

SDO	Contribution	Resolution	Amendment	Specification artifact
3GPP (this paper)	TDoc, LS	Agreement / Conclusion / Working Assumption	Change Request (CR)	TS (normative); TR (informative)
IETF (analogy)	Internet-Draft	WG consensus / IESG approval	bis-draft / RFC errata [9]	RFC
IEEE 802 (analogy)	Submission, Comment	Letter Ballot, motion	PAR amendment, sponsor ballot	IEEE Std

Hypothetical analogy only; cross-SDO empirical validation is left to future work.

3.3 Concepts an Ontology Must Capture

Three features shape the ontology: **scale** (10^4–10^5 TDocs per WG; cf. Table 6); **document lineage** (TDoc→Resolution→CR→Section→Spec demands first-class types and directed links per document); and **cross-group dependen-**

cies (Liaison Statements as directed artefacts, not free text). These requirements inform the four lifecycle concerns and supporting class clusters described in §4.

4 SPECTRA Ontology Design

4.1 Competency Question Analysis

SPECTRA is grounded in 137 competency questions collected across five development phases by practicing 3GPP RAN delegates within the authors' organization (external-SME validation deferred, see §8). Every property exercised by the 137 CQs maps to a field publicly available on 3GPP TDoc cover sheets, chairman-note resolutions, or per-meeting `TDOC_List.xlsx`—the dataset is structurally grounded in externally verifiable 3GPP records rather than internal-only metadata. The **137-CQ dataset (*SpectraCQ v1.0*)** with verbatim company names matching the public 3GPP `TDOC_List.xlsx` (full per-tier policy in Appendix F) — English text, Cypher query specifications validated on the internal KG, phase, category, schema area, and RAN1 per-CQ verdict — is shipped at `cqs/spectra_cq_v1.0/` under CC-BY 4.0; the per-CQ PASS verdict is carried into the public `validation/cq_results.json` reproduced by `tests/verify_release.py`, and a representative subset runs against the bundled synthetic and metadata-only mini examples.

CQs are distributed across the development phases as: P1 TDoc metadata (25, e.g. "What TDocs did Huawei submit?"), P2 Meeting resolutions (34, "What agreements on Agenda Item Z?"), P3 TS structure (45, "Which CRs modified section S?"), P4 CR documents (15, "What is the reason for change of CR C?"), P5 Technical reports (18, "Which TRs have impacted TS 38.211?"), totalling 137.

4.2 Core Schema

Based on the CQ analysis, we designed SPECTRA following the Ontology 101 methodology [12]. The ontology is encoded in OWL 2 [8] and comprises 26 classes, 53 object properties (OPs), and 81 data properties (DPs). These classes operationalize the four lifecycle concerns introduced in §1 (contributions, resolutions, specifications, cross-group communication) plus organizational and auxiliary-content clusters needed for joins and extracted evidence (Figure 2).

Document: `Tdoc` with four subclasses (`CR`, `LS`, `Summary`, `Session Notes`); `CR` → `Spec` via `modifies` and `CR` → `Section` via `modifiesSection`; `LS` → `WorkingGroup` via `sentTo`/`originatedFrom`. **Resolution**: subclasses `Agreement`, `Conclusion`, `WorkingAssumption` (with `promotedTo` linking a `WorkingAssumption` to a later `Agreement`, often after cross-WG or cross-meeting dependencies are resolved). **Specification**: `Spec` (TS) and `Section` forming a tree via `hasSubSection`/`parentSection` (irreflexive, asymmetric); `TechnicalReport` links to influenced Specs through a reified `TR Impact` so a single TR can affect multiple Specs with per-impact metadata. **Organizational**: `Meeting`, `Company` (⊑ `foaf:Organization`), `Contact` (⊑ `foaf:Person`), `WorkItem`, `AgendaItem`, `Release`, `WorkingGroup`. **Auxiliary content**: `Figure`, `Table`, `Chart`, `TSTable`, `TSFigure` capture structured content

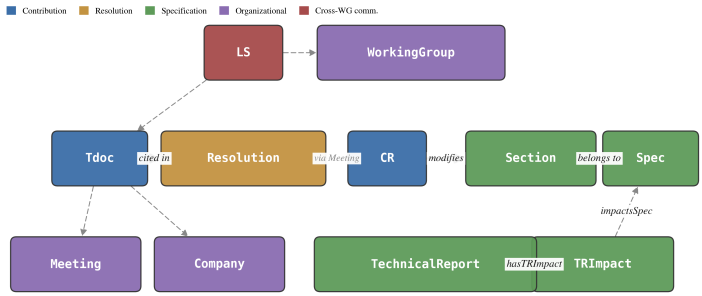

Fig. 2. SPECTRA core schema (reader-facing labels). **Central row**: traceability spine `Tdoc` → `Resolution` → `CR` → `Section` → `Spec`. Formal OPs are `references` (Resolution → Tdoc, drawn reversed), `modifiesSection` (CR → Section), and `belongsToSpec` (Section → Spec). The Resolution → CR step is drawn lightly because no direct OP exists; the connection is reconstructed by joining shared `Meeting`, `AgendaItem`, and `CRPack` context. **Dashed grey arrows** carry secondary OPs (LS → WG via `sentTo`; Tdoc → {Meeting, Company}) and one class-hierarchy edge (LS $\sqsubseteq$ Tdoc). The TR → TRImpact → Spec reified chain is shown explicitly. Section hierarchy edges (`hasSubSection`/`parentSection`, irreflexive + asymmetric) and the four intra-section / TR cross-reference irreflexive OPs (`referencesSection`/`referencedBySection`, `referencesTR`/`referencedByTR`) are realized inside the Section/TR sub-trees.

extracted from contributions and specifications for downstream traceability tools (figure-to-section linkage, table-to-CR provenance) even though they sit off the main lifecycle spine; per Table 6 they are non-trivially populated (`Table`=1,441 instances).

Key Design Decisions. **D1:** Document types are OWL subclasses (not property values) so that type-specific properties can be declared per subclass. **D2:** 20 object properties carry `owl:FunctionalProperty` for an *at-most-one* contract (e.g., a Section `belongsToSpec` at most one Spec; each `TRImpact` points to at most one impacted Spec, while a single TR may have multiple `TRImpact` nodes); the loader enforces these at ingest time. Re-tableable links (`presentedAt`) and links that are legitimately multi-valued, implicit, or source-dependent (`modifies`, `originatedFrom`/`sentTo`) are deliberately not declared functional — see Appendix G. **D3:** The spine (Figure 2) is realized by three direct OPs (`references`, `modifiesSection`, `belongsToSpec`); the Resolution → CR step is an implementation-level join combining `madeAt` (Resolution → Meeting), `presentedAt` (CR → Meeting), and the shared `AgendaItem` (via `resolutionBelongsTo` from Resolution and `belongsTo` from CR/Tdoc); for TSG CR-pack approval flows, `belongsToCRPack` (CR → CRPack) augments the join via the literal `tsgMeeting` attribute on CRPack—no single OP, by design, so each declared property keeps a single, verifiable semantic. **D4:** TR-to-Spec impact is reified as `TRImpact` rather than a direct OP because a single TR study typically affects multiple Sections of multiple Specs with a per-impact `impactType` (one of *NewTS, NewSection, ExtendSection, NoChange*; range `xsd:string` in OWL with the closed value set enforced by

SHACL `sh:in` in `shapes/spectra-core.shacl.ttl`) carried alongside the impacted section/clause; a direct OP would lose that per-impact context. **D5:** An *optional* PROV-O alignment ships as 6 `rdfs:subClassOf` axioms (`Resolution` ⊑ `prov:Activity`; `Tdoc` ⊑ `prov:Entity`; `Company` ⊑ `prov:Agent`, `prov:Organization`; `Contact` ⊑ `prov:Agent`, `prov:Person`); we deliberately avoid `owl:imports` so consumers without a PROV reasoner pay no cost while reusers can opt in for cross-domain provenance queries.

4.3 Reuse of Existing Vocabularies

SPECTRA reuses established Semantic Web vocabularies where they naturally apply (Table 3). Domain-specific classes are introduced only where no suitable external term exists.

Table 3. Reuse of external vocabularies in SPECTRA, by slot.

Vocabulary	Slot	Terms used
Dublin Core [5]	Metadata	`dc:title`, `dc:description`, `dc:creator`, `dc:date`, `dc:rights`
DCTERMS	Licensing	`dcterms:license`
FOAF [3]	Agent class	`foaf:Person` (⊇ `Contact`), `foaf:Organization` (⊇ `Company`)
OWL 2	Axioms	`Functional/InverseFunctional/Irreflexive/Asymmetric Property`, `inverseOf`
RDFS	Hierarchy	`subClassOf, label, comment`
PROV-O [10] (optional)	Provenance	6 `rdfs:subClassOf` alignment axioms (Resolution⊑Activity; Tdoc⊑Entity; Company/Contact⊑Agent; Company⊑Organization; Contact⊑Person)

We considered but did *not* adopt **ORG** (Organization/Unit/Membership beyond CQ needs), **SKOS** (hierarchical term taxonomies rather than typed entities), or **DCAT/VoID** (dataset description). The optional PROV-O alignment ships as 6 `rdfs:subClassOf` axioms in the OWL file but `owl:imports` is not asserted, so consumers can opt in without pulling the full PROV-O TBox; SPECTRA itself supplies the 3GPP-specific roles (TDoc, CR, LS, Agreement, Working Assumption, AgendaItem, CRPack, TS Section).

4.4 Constraints and Axioms

SPECTRA includes 20 functional + 2 inverse-functional (`hasCR`, `has TRImpact`) properties, 15 inverse pairs, 6 irreflexive (section hierarchy, intra-section + TR cross-references), and 2 asymmetric (section hierarchy) declarations. SHACL shapes (`shapes/spectra-core.shacl.ttl`) capture cardinality and range portably—e.g., a `spectra:Tdoc` is required to carry exactly one `tdocNumber` (`sh:minCount 1; sh:maxCount 1`) and a `Section` carries at most one `belongsToSpec` link (`sh:maxCount 1`). Lifecycle links with multi-valued, implicit, or source-dependent behavior (`presentedAt` under re-tabling, `originatedFrom/sentTo` for cross-WG LSs, `modifies` for paired CRs) are deliberately left without strict cardinality bounds; their data-quality conventions are documented in Appendix G.

5 Cumulative CQ Quality Assurance

This section describes the cumulative CQ regression practice that produced the released ontology — a QA discipline complementary to established CQ-driven methodologies (LOT, SAMOD, NeOn) and presented here as a process-engineering contribution rather than a replacement methodology: each new phase must leave every previously validated competency intact, so that schema refinements are caught at the point of introduction rather than at release.

5.1 Five-Phase Progressive Construction

SPECTRA was developed through five incremental phases — **P1** TDoc cover-sheet metadata, **P2** meeting-level resolutions (Agreement / Conclusion / WorkingAssumption), **P3** TS section structure, **P4** CR documents and CR-pack semantics, **P5** TR impact on TS — each extending the ontology while re-executing the full CQ suite accumulated from prior phases. The class/OP/DP trajectory (classes / object properties / data properties) is 11/14/31 at P1+P2 → 17/23/52 at P3 → 24/43/78 at P4 → 26/51/87 at P5, with v1.0.0 ending at 26/53/81 after refactoring identified through cross-phase integrity verification (Figure 3).

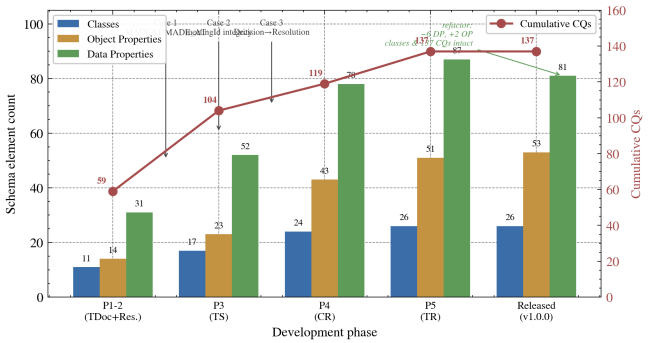

Fig. 3. Schema growth across the five-phase development of SPECTRA, measured from git snapshots at phase-closing commits. The rightmost "Released" bar is the public v1.0.0 state after a cross-phase refactoring: e.g., `hasRelatedTR` added for explicit `WorkItem`–TR linkage (TS–TR impact remains via reified `TRImpact`); unused Meeting attributes removed; net −6 DPs and +2 OPs, with classes and 137 CQs unchanged. Cumulative CQs are on the right axis.

5.2 Quality-Assurance Cases

Three representative cases (traceable in git history) illustrate what single-phase validation would have missed: **Case 1** (P2→P3) harmonized loader-side Cypher relationship labels (`MADEAT` vs `MADE_AT`, both mapping to `madeAt`) — multiple Phase-1–2 CQs would have silently regressed. **Case 2** (P3 audit) patched meeting-ID referential-integrity gaps that broke `Tdoc` → `Meeting` → `Resolution` joins. **Case 3** (P3→P4) resolved three

schema refinements: `releaseId` removed, `hasCR` declared inverse-functional, and `Decision` renamed to `Resolution` (cumulative CQ counts at the four transitions: 59, 104, 119, 137).

5.3 Cumulative Regression Verification

At each transition, all CQs from Phases 1–N are re-validated against the Phase-N schema over the internally instantiated KG — this is an internal protocol; the public artifact ships the final-state RAN1 verdict (`validation/cq_results.json`, 137/137 PASS) rather than a per-transition matrix. The 137 CQs accumulated by Phase 5 all return non-empty results manually judged correct by the eliciting engineers; after the QA cases above, our internal records did not surface further regressions at subsequent transitions.

6 Evaluation

We organise the evaluation around six dimensions: (E1) requirements coverage by CQs; (E2) structural soundness; (E3) instantiation scale; (E4) cross-WG reuse; (E5) public reusability; (E6) demonstrated utility against SOTA LLM baselines. Release-scope and licensing rationale is in §7.

6.1 E1 — Requirements coverage by CQs

All 137 CQs (137/137 PASS on RAN1; per-phase 25/34/45/15/18) return non-empty results judged plausible by the eliciting engineers; `verify_release.py` verifies the released verdict files (not re-execution against the operational KG, threat acknowledged in §8). Structural and CQ-coverage claims are publicly re-derivable via `tests/verify_release.py`. The 137 CQs exercise 20 of 26 classes and 31 of 53 object properties. The 6 unexercised classes are `LS` plus Chart/Figure/Table/Summary/SessionNotes; LS-typed TDocs are exercised through their `Tdoc` superclass via three Phase-1 CQs covering `originatedFrom`/`sentTo`/`replyTo`/`ccTo`, and full LS chains run via use scenario S2 over `examples/process_kg/` (26,791 LSs across the RAN1–RAN5 union). `modifiesSection` and `promotedTo` are not directly exercised by the 137-CQ regression set but are covered by use scenarios S1/S6 and S4 respectively (Table 8).

6.2 E2 — Structural quality

We report structural metrics of the released ontology (Table 4) and the findings of an automated pitfall scan.

OOPS! [17] reports findings in 0 Critical, 2 Important, and 2 Minor pitfall categories (Table 5). The two Important categories have distinct rationales: P34 is a direct consequence of FOAF reuse, while P10 reflects a deferred `owl:disjointWith` decision because none of the 137 CQs depend on disjointness reasoning. The release ships PyLODE HTML and a SHACL file (`spectra-core.shacl.ttl`) validated against the bundled synthetic instantiation via pySHACL; the larger process KG is checked in Appendix G.

Table 4. Ontology structural metrics. The reused-vocabulary count excludes the 6 optional PROV-O alignment axioms.

Metric	Value
Classes	26
Object Properties	53
Data Properties	81
Functional Properties	20
Inverse Functional Properties	2
Inverse Property Pairs	15
Irreflexive Properties	6
Asymmetric Properties	2
Core reused vocabulary terms 8 (5 DC + 1 DCTERMS + 2 FOAF)	

Table 5. OOPS! pitfall scan summary; design rationale shown for each finding.

Code	Severity	#	Design rationale
P34	Important	2	FOAF `Person`/`Organization` reused without redeclaration
P10	Important	1	`owl:disjointWith` on Tdoc subclasses deferred (CQs do not rely on it)
P08	Minor	2	Annotations omitted on reused FOAF terms
P13	Minor	23	`owl:inverseOf` declared only on bidirectionally queried pairs

6.3 E3 — Large-scale instantiation evidence

To provide scale evidence from the operational instantiation, we instantiated SPECTRA against RAN1 data. The graph materializes 123,677 unique RAN1 TDocs from 60 meetings; the released `ran1_tdoc_metadata.ttl` preserves 123,678 source metadata records because 207 TDocs (0.17%) carry multiple `presentedAt` edges under re-tabling and one re-tabled record shares its identifier with an earlier record (Appendix G). All associated Resolutions, CRs, LSs, Specs, and TRs are reported in Table 6 (snapshot dated April–May 2026; per-file timestamps in `validation/`). Relation quality (denominators in `validation/ran1_relation_integrity.json`): `submittedBy` referential integrity 99.36% (1,000 missing of 155,525 Tdoc-Company source records, one record per multi-author TDoc per submitting company); `presentedAt`/`madeAt` loader-enforced 100% (per Tdoc/Resolution instance); `references`/`modifies` `Section` machine-resolvable linkage coverage 51.55%/34.66% (over source-cited reference rows; partly reflects 3GPP citation granularity, with remaining gaps combining parser limits and free-text citations rather than schema gaps). Aggregate counts, metadata-only derivatives, and per-WG body-text dumps are all in v1.0.0 (`validation/`, `examples/process_kg/`, `kg/per_wg/`; §7).

6.4 E4 — Cross-WG generality of the schema

A schema designed for one WG can fail silently when applied to a sibling WG with different document conventions. We applied SPECTRA — originally designed from RAN1 CQs — to RAN2–RAN5 without redesign and measured how many *additional* schema elements each WG needed (Table 7).

The 26 classes cover every document-type distinction the per-WG loaders produced (**zero** additional classes). One object property (`REFERENCES_SPEC`) and two data properties are added (RAN3 `summaryId`, RAN5 `discussionText`; the 22 apparent DP extensions decompose into 20 loader-local attributes plus

Table 6. Per-class instance counts in the RAN1 instantiation (operational snapshot, April–May 2026; `validation/ran1_instance_counts.json`). `WorkingGroup`=116 covers RAN1–RAN5 plus other TSGs and external bodies cited via `originatedFrom`/`sentTo` on LS cover sheets.

Class	#	Class	#	Class	#
Tdoc	123,677	Resolution	24,710	Section	1,003
CR	10,715	Agreement	21,058	TSTable	680
LS	6,343	Conclusion	2,703	TSFigure	13
Summary	3,468	WorkingAssump.	949	CRPack	501
SessionNotes	678	Meeting	60	TechnicalReport	26
Spec	83	Company	222	TRImpact	29
Release	13	Contact	1,000	Figure	120
WorkingGroup	116	WorkItem	423	Table	1,441
AgendaItem	6,439			Chart	1

Table 7. Schema diff of RAN2–RAN5 instantiations against the RAN1-designed SPECTRA baseline. "WG-specific extensions" counts schema elements observed in a non-RAN1 WG graph beyond the original RAN1 baseline (union over RAN2–RAN5); they are reported as extension candidates rather than evidence of class-level redesign. Extension rate is computed against the corresponding released-SPECTRA element count (1 of 53 OPs, 2 of 81 DPs).

Schema element	In released SPECTRA	WG-specific extensions	Extension rate
Classes	26	0	0.0%
Object Properties	53	1	1.9%
Data Properties	81	2 †	2.5%

† excluding 20 loader-local properties (17 underscore-prefixed mirror slots and 3 internal scaffolding properties).

these two). Total additional schema surface across four sibling WGs: 1 OP and 2 DPs on top of 160 elements — schema-level reuse evidence within the released RAN1–RAN5 snapshots of the RAN TSG, not a universal transferability claim.

Conversely, 6 of the 26 RAN1 classes are absent from at least one sibling WG's instantiation. **Per-WG class coverage** (Figure 4; released in `validation/per_wg_class_coverage.json`): RAN1 26/26 (100%), RAN2 22/26 (84.6%), RAN3 23/26 (88.5%), RAN4 24/26 (92.3%), RAN5 20/26 (76.9%); in the released RAN5 conformance-testing snapshot, Agreement, Chart, Figure, SessionNotes, Summary, and WorkingAssumption are not populated. These absences reflect released-snapshot and loader scope, not schema gaps.

6.5 E5 — Reusability evidence via public artifacts

To illustrate what the ontology is for, Table 8 lists seven curated competency-question scenarios. Each pairs a natural-language question with the schema fragment it exercises and the role it plays in the standardization workflow; S2–S3 cite cardinalities measured on the deployed RAN1–RAN5 KGs.

SpectraCQ v1.0 ships at `cqs/spectra_cq_v1.0/`; cross-WG reuse is evidenced at schema level (`validation/cross_wg_schema_diff.json`) rather than per-WG CQ re-execution.

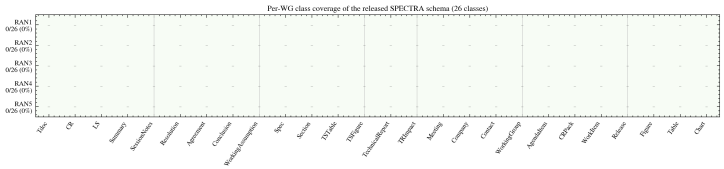

Fig. 4. Per-WG class coverage of the released SPECTRA schema across the five RAN Working Groups. Filled cells (▪) indicate the class is instantiated in that WG; dashes mark absences. Coverage totals are shown beside each WG. RAN5's lower coverage reflects its conformance-testing scope (no Resolution/auxiliary content), not a schema gap.

Table 8. Use scenarios S1–S7 supported by SPECTRA. Schema fragments cite OWL property names from the released TTL.

Scenario	Question	Schema fragment / signal
S1 Multi-hop trace	TDoc lineage behind current text of TS 38.214 §5.1.3	`Section` ← `CR` → `Meeting` ← `Resolution` → `Tdoc`; uses `modifiesSection`, `presentedAt`, `madeAt`, `references`
S2 Cross-WG LS impact	RAN2 contributions in meeting M responding to RAN1 LSs	`originatedFrom`, `sentTo`, `replyIn`; deployed: 3,644 LSs RAN1→RAN2 (top recipient); 1,450 received in RAN2 KG, peak RAN2#118-e (62)
S3 Release CR analytics	CRs against Release 18 by Huawei	`targetRelease`, `modifies`, `submittedBy`; deployed: TS 38.300 has 2,039 RAN2 + 270 RAN3 CRs
S4 WA promotion lineage	WAs promoted to Agreements + meeting span	`WorkingAssumption` $\xrightarrow{\texttt{promotedTo}}$ `Agreement`, joined on `madeAt`
S5 TR→TS impact	TS sections affected by TR 38.913 conclusions	reified `TRImpact` via `hasTRImpact`, `impacts` `Section` (1-TR-to-many-TS)
S6 Per-company profile	TS 38.331 sections modified by CRs from Samsung over 4 meetings	`submittedBy`, `presentedAt`, `modifiesSection`, `belongsToSpec`
S7 Work-Item lineage	TDocs/CRs associated with agreements under WI `NR_e MIMO-Core`	`relatedToWorkItem`, `madeAt`, `references`, `hasCR`

6.6 E6 — Demonstrated utility against SOTA LLM baselines

To probe whether SPECTRA-grounded retrieval offers measurable value over generic LLM access to 3GPP knowledge, we compared (i) a SPECTRA-conformant Neo4j+Qdrant RAG on the released schema, (ii) GPT, and (iii) Claude on four cross-WG practitioner questions (Rel-16 Type-II codebook, TCI-state Rel-15 to Rel-20, Beam Failure Detection/Recovery, Rel-18 Layer-1/Layer-2 Triggered Mobility). Each answer was scored against authority sources (3gpp.org, ETSI TS, IEEE Xplore, sharetechnote, Ofinno) on a five-axis rubric and verified for hallucinations (Table 9).

SPECTRA-grounded retrieval wins four of five axes; Claude leads on Coverage (4.58 vs 3.95) due to ASN.1 IE bodies not yet chunked by SPECTRA's loader (roadmap). Citation Integrity gap (+2.45 over Claude) reflects the structural advantage of KG-grounded answers. Detailed evaluation, retrieval logs, and

Table 9. Q&A quality on four cross-WG questions (5-axis rubric, 0–5 scale, 4Q averages). Hallucination row reports detected unsupported quotes/codes/numbers across all four questions.

Axis (4Q avg)	SPECTRA RAG	GPT	Claude
A1 Accuracy (vs authority)	**4.55**	3.65	3.75
A2 Coverage (depth/breadth)	3.95	3.78	**4.58**
A3 Citation Integrity	**4.83**	1.28	2.38
A4 Hallucination Control	**4.85**	3.63	3.08
A5 Cross-Doc Linkage	**4.58**	3.95	4.53
Composite	**4.55**	3.28	3.65
Hallucination instances	**0**	1	∼11

rubric: `usecase/`. This is a single-evaluator pilot; broader evaluation is future work.

7 Availability, Sustainability, and FAIRness

The **v1.0.0 release** ships SPECTRA-authored components (ontology, SHACL shapes, *SpectraCQ v1.0*, queries, examples, validation JSONs, `verify_release.py`, sanitized parsing pipeline) under CC-BY 4.0; per-WG body-text KGs (RAN1–RAN5) are 3GPP-derived literals retained with explicit 3GPP source attribution, following the academic redistribution practice of [11,23]. The ontology IRI is `https://w3id.org/spectra` (w3id PR submitted); `validation_manifest.md` maps each quantitative claim to its supporting file.

FAIR-R alignment. **F:** persistent IRI `https://w3id.org/spectra` (w3id PR submitted); Zenodo DOI to be minted at v1.0.0 release; LOV submission planned. **A:** ontology, docs, CQs, examples, body-text KGs, pipeline, and validation files downloadable from the public GitHub repo without authentication. **I:** OWL 2 Turtle (RDFLib-loadable); SHACL Core (`shapes/spectra-core.shacl.ttl`) validated under pyshacl with no SHACL-AF; reuses DC/DCTERMS/FOAF; optional PROV-O alignment. **R:** CC-BY 4.0; deterministic `verify_release.py` runs 9 reproducibility checks (25 sub-checks total, all passing); per-claim mapping in Appendix E.

Sustainability. SPECTRA is the semantic layer of a Samsung System LSI standards-engineering KG with per-meeting incremental updates passing the cumulative CQ protocol; annual snapshots align to the RAN plenary cadence. A static scan (`validation/cypher_to_sparql_portability.json`) finds 136 of 137 Cypher queries use only SPARQL-equivalent constructs (illustrated by the 6 bundled SPARQL examples).

8 Conclusion

We presented **SPECTRA**, a reusable OWL ontology for the 3GPP RAN standardization document lifecycle, grounded in 137 CQs (standalone *SpectraCQ v1.0*) and exercised through five-phase cumulative QA. Cross-WG reuse: only

three extension candidates (one OP, two DPs) on RAN2–RAN5. **Threats:** CQs verified by eliciting engineers (no external panel); adoption confined to the authors' organisation; cross-SDO transferability (IETF/IEEE) is design intent. **Future:** VectorDB embeddings, PROV-O process OPs, TSG SA/CT extension, LLM-assisted CQ generation [19], external evaluation.

Resource Availability Statement. `release_package/` ships v1.0.0 under CC-BY 4.0 (SPECTRA-authored components); per-WG body-text KGs (`kg/per_wg/`) are 3GPP-derived literals under explicit 3GPP source attribution, following [11,23]. Repo `https://github.com/spectra-ontology/spec-trace`; persistent URI `https://w3id.org/spectra` (w3id PR submitted); Zenodo DOI to be minted at v1.0.0 release. Cite as Choi, Lee, Ahn (2026), *SPECTRA v1.0.0*, Zenodo. **Acknowledgments.** We thank colleagues at the System LSI Business of Samsung Electronics for domain feedback.

Declaration of use of Generative AI

In accordance with the ISWC 2026 Generative AI policy, we disclose the following uses of generative AI tools during the preparation of this Work and the accompanying resource. *Generative AI is not listed as an author.*

- **Manuscript text**: Anthropic Claude Code and OpenAI Codex were utilized to generate sections of this Work including English-prose language editing, paragraph restructuring, and consistency checks across sections (terminology and abbreviation harmonization, citation cross-references).
- **Code and scripts**: Generative AI assisted in scaffolding utility scripts in the release package (`tests/verify_release.py`, structural-metric reproducibility scripts, Cypher/SPARQL templates) and the per-claim validation manifest. All released code was reviewed, executed, and validated by the authors before release; `tests/verify_release.py` runs deterministically.
- **Figure source**: matplotlib/TikZ source for Figures 1, 2, 3 was partly scaffolded with AI assistance and visually reviewed by the authors; all numerical content shown in figures originates from `validation/*.json`.
- **Tables**: all data values in Tables 1–8 were authored manually from validation JSON evidence and the released artifacts; AI assistance was limited to LaTeX formatting suggestions.
- **Citations and bibliography**: AI assistance in identifying related work was double-checked by the authors against the cited PDFs; final `.bib` entries are author-curated.
- **Downstream CQ validation (described in §7)**: An LLM (OpenRouter Gemini-2.5-flash) was used for natural-language-CQ-to-Cypher translation suggestions and answer-quality scoring during internal CQ-design iterations. *All released Cypher queries and the 137-PASS verdict were manually reviewed by the eliciting engineers; no LLM output enters the KG-population path or the deterministic verification script.*

– **Not used for**: ontology design choices (class/property selection, axiomatization), competency-question elicitation, schema-diff judgments, SHACL-shape design, evaluation results, or substantive scientific content.

The authors take full responsibility for all content of this Work and the released resource.

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Appendix

A Ontology TTL and SHACL Excerpts

Ontology TTL (illustrative excerpt). Figure 5 shows the `spectra:Tdoc` class declaration with one functional and one non-functional OP, adapted from `ontology/spectra.ttl` (English labels and the explanatory `#` comments are paper-side annotations; the full TTL with all comments and the PROV-O block lives at the cited path).

SHACL shapes (illustrative excerpt). The full `shapes/spectra-core.shacl.ttl` declares 8 `sh:NodeShape`s covering the lifecycle classes. Figure 6 shows the `Tdoc` and `CR` shapes excerpted from the released file (the file uses the local `tdoc:` prefix bound to `https://w3id.org/spectra#`; we use `spectra:` below for readability).

B Sample SPARQL Queries

Figures 7 and 8 render two of the seven use scenarios from §6 as portable SPARQL. The Cypher equivalents distributed with SpectraCQ v1.0 follow the

```
spectra:Tdoc a owl:Class ;
    rdfs:subClassOf prov:Entity ;
    rdfs:label "Technical Document"@en .

spectra:submittedBy a owl:ObjectProperty ;
    rdfs:comment "Multiple companies can co-submit." ;
    rdfs:domain spectra:Tdoc ; rdfs:range spectra:Company .

spectra:hasContact a owl:ObjectProperty , owl:FunctionalProperty ;
    rdfs:domain spectra:Tdoc ; rdfs:range spectra:Contact .

spectra:presentedAt a owl:ObjectProperty ;
    rdfs:domain spectra:Tdoc ; rdfs:range spectra:Meeting .
    # Not declared owl:FunctionalProperty: re-tabling admits
    # multiple Meeting values per Tdoc (Appendix \ref{app:dq}).
```

Fig. 5. Excerpt from `ontology/spectra.ttl`. `hasContact` is functional (a Tdoc has at most one responsible Contact); `submittedBy` and `presentedAt` are intentionally non-functional to admit co-submitting companies and re-tabled TDocs respectively.

```
spectra:TdocShape a sh:NodeShape ;
    sh:targetClass spectra:Tdoc ;
    sh:property [ sh:path spectra:tdocNumber ; sh:datatype xsd:string ;
                  sh:minCount 1 ; sh:maxCount 1 ] ;
    sh:property [ sh:path spectra:presentedAt ; sh:class spectra:Meeting ] ;
    sh:property [ sh:path spectra:targetRelease ; sh:class spectra:Release ;
                  sh:maxCount 1 ] .

spectra:CRShape a sh:NodeShape ;
    sh:targetClass spectra:CR ;
    sh:property [ sh:path spectra:modifies ; sh:class spectra:Spec ] ;
    sh:property [ sh:path spectra:belongsToCRPack ; sh:class spectra:CRPack ;
                  sh:maxCount 1 ] .
```

Fig. 6. Excerpt from `shapes/spectra-core.shacl.ttl`. `tdocNumber` is required and unique per Tdoc; `targetRelease` is bounded above (a TDoc targets at most one Release). `presentedAt` carries no `sh:maxCount` because ∼0.17% of TDocs are re-tabled across meetings under the same identifier (Appendix G); `modifies` is multi-valued for paired stage-2/stage-3 CRs.

same join structure but use Neo4j syntax. Identifier prefixes are abbreviated for the figure.

C Per-WG Class Coverage

Table 10 expands the per-WG class-coverage summary cited in §6.4; the full per-class breakdown lives in `validation/per_wg_class_coverage.json`. Absences reflect the released RAN-WG snapshot and loader scope rather than schema gaps: e.g., the released RAN5 snapshot does not instantiate `Agreement`, `WorkingAssumption`, `Summary`, and `SessionNotes`.

Table 10. Per-WG class coverage of the released SPECTRA schema, showing classes absent in each non-RAN1 WG instantiation.

WG	Coverage	Classes absent (compared with RAN1's 26)
RAN1	26/26 (100%)	—
RAN2	22/26 (84.6%)	Chart, Figure, SessionNotes, Summary
RAN3	23/26 (88.5%)	Chart, Figure, SessionNotes
RAN4	24/26 (92.3%)	SessionNotes, Summary
RAN5	20/26 (76.9%)	Agreement, Chart, Figure, SessionNotes, Summary, WorkingAssumption

```
# S1: TDoc lineage behind current text of TS 38.214 §5.1.3
SELECT ?tdoc ?meeting ?date WHERE {
  ?section spectra:belongsToSpec spectra:TS_38.214 ;
           spectra:sectionNumber "5.1.3" .
  ?cr      spectra:modifiesSection ?section ;
           spectra:presentedAt    ?meeting .
  ?res     spectra:madeAt         ?meeting ;
           spectra:references     ?tdoc .
  ?meeting spectra:meetingStartDate ?date .
} ORDER BY DESC(?date)
```

Fig. 7. S1 multi-hop traceability in SPARQL (simplified for the listing; the production Cypher additionally constrains `AgendaItem`/`CRPack` context per the spine-join rationale in §4, and folds the `Agreement`/`WorkingAssumption`-promotion `OPTIONAL` branches into the single `spectra:references` step shown here). The Resolution–CR join is implementation-level via shared `Meeting`.

```
# S5: TS sections affected by TR 38.913 study, with impact type
SELECT ?spec ?section ?impactType WHERE {
  ?tr     a spectra:TechnicalReport ;
          spectra:trNumber "38.913" ;
          spectra:hasTRImpact ?ti .
  ?ti     spectra:impactType   ?impactType ;
          spectra:impactsSpec  ?spec ;
          spectra:impactsSection ?section .
}
```

Fig. 8. S5 TR-to-TS impact propagation in SPARQL (TR-anchored direction; the released SpectraCQ S5 query file uses the inverse TS-anchored direction via `impactOfTR`, which the ontology supports symmetrically). Each `TRImpact` carries an `impactType` from the closed set {NewTS, NewSection, ExtendSection, NoChange}.

D Cumulative CQ Regression Cases (Extended)

This appendix expands the three QA cases summarised in §5. Each entry indicates the phase transition, the schema or naming change introduced, the class of previously-passing CQs that would have silently regressed without the cumulative re-execution, and the resolution. Per-case CQ counts are not extracted from the development history; the cumulative pass record (`validation/cq_results.json`) reports the post-fix steady state.

- **Case 1 (Phase-2→3, naming inconsistency `MADEAT` vs `MADE_AT`).** Three relationship labels and three property names were harmonized. Multiple Phase-1–2 CQs that filtered on the relationship name returned empty after the introduction of Phase-3 schema until the naming was unified across loaders and ontology.
- **Case 2 (Phase-3 audit, meeting-ID referential integrity).** A subset of Phase-1–2 CQs that joined `Tdoc` → `Meeting` → `Resolution` regressed when meeting identifiers used inconsistent capitalisations across CR/Resolution loaders. Three targeted patches resolved the gap without changing the schema.
- **Case 3 (Phase-3→4, three schema refinements).** `releaseId` was removed as redundant once `targetRelease` was declared functional; `hasCR` was declared `owl:InverseFunctionalProperty` so that a CR could be unambiguously located from its CR-pack; and the base class `Decision` was renamed to `Resolution` for terminological alignment with 3GPP chairman-

note vocabulary. A further set of Phase-1–2 CQs that referenced the renamed class would have silently regressed.

Cumulative CQ counts at the four transitions were 59, 104, 119, and 137. Per-CQ pass/fail is recorded in `validation/cq_results.json`; the regressions surfaced by cumulative re-execution would not have been detected by single-phase validation before release.

E Reproducibility Checklist (Mapping to Release Files)

Table 11 expands the Reproducibility paragraph in §7 into a per-claim mapping. Each row links a public claim made in the paper to the file(s) under `release_package/` that support it and to the `verify_release.py` check that re-derives it. All nine checks pass on the v1.0.0 release (25/25 sub-checks).

Table 11. Public-claim → artifact → `verify_release.py` mapping.

Claim	Artifact	Check
Ontology has 887 triples	`ontology/spectra.ttl`	`verify_release.py §1`
Bundled instantiation snippet conforms to SHACL	`examples/instantiation_snippet.ttl`, `shapes/spectra-core.shacl.ttl`	`verify_release.py §2`
Process-KG union conforms to SHACL (∼2.44M triples, Appendix G)	`examples/process_kg/*.ttl`, `shapes/spectra-core.shacl.ttl`	`verify_release.py §2b`
End-to-end SPARQL example returns the documented row	`examples/end_to_end/`	`verify_release.py §3`
SpectraCQ ships 137 CQ + 137 Cypher with 25/34/45/15/18 phase distribution and 137-PASS RAN1 verdict	`cqs/spectra_cq_v1.0/`	`verify_release.py §4`
Structural metrics (26/53/81/20/2/15/6/2 + 6 PROV-O axioms)	`validation/structural_metrics.json`	`verify_release.py §5`
All validation manifest references resolve	`validation/validation_manifest.md`	`verify_release.py §6`
SpectraCQ text and Cypher contain no real company names	`cqs/spectra_cq_v1.0/`	`verify_release.py §7`

The `examples/process_kg/` directory (LS/CR routing across RAN1–RAN5 + RAN1 TDoc structural metadata, body content stripped) is verified via SHACL conformance against `shapes/spectra-core.shacl.ttl`. Both `examples/process_kg/` and `cqs/spectra_cq_v1.0/` retain verbatim company names matching the public 3GPP `TDOC_List.xlsx`.

F Artifact Treatment, Attribution, and Redistribution

The release bundles four artifact tiers governed by different rules (Table 12). The treatment reflects *what the data is*: 3GPP-public metadata (including company names from `TDOC_List.xlsx`) is redistributed verbatim, illustrative TTL

uses synthetic identifiers, and 3GPP-derived body text is retained under explicit 3GPP attribution.

Table 12. Per-tier treatment, license/attribution, and rationale.

Artifact	Treatment	License / attribution	Rationale
`examples/process_kg/` (routing edges + RAN1 TDoc metadata)	verbatim metadata	CC-BY 4.0	Redistributes metadata already public on 3GPP's per-meeting `TDOC_List.xlsx` and TDoc cover pages; redacting would discard recoverable information.
`cqs/spectra_cq_v1.0/` (NL questions, Cypher)	verbatim company names	CC-BY 4.0	Released as an executable CQ-to-query benchmark; company names match `TDOC_List.xlsx` so queries return real results on any 3GPP-conformant KG.
`examples/real_world_mini/`, `examples/end_to_end/`	synthetic identifiers	CC-BY 4.0	Templates for instantiation; no real-world data is implied.
Body content (CR/TR/TS text, `discussionText` *values*)	verbatim	3GPP-derived; explicit 3GPP source attribution	Sentence-level rendered text retained with explicit 3GPP attribution, following the academic redistribution practice established by [11,23]. Original PDFs remain freely accessible at `https://www.3gpp.org/ftp/`.

Company names. Both process-KG metadata and SpectraCQ retain verbatim company names matching the public 3GPP per-meeting `TDOC_List.xlsx` and TDoc cover pages. No anonymization is applied because the names are already public 3GPP metadata. The `Contact` class instances populated in our internal operational KG are excluded from the public release; the released TTLs contain no personal correspondence from document body text.

Two-tier licence boundary. SPECTRA-authored components (the OWL ontology, SHACL shapes, PyLODE documentation, SpectraCQ v1.0, sample SPARQL/Cypher queries, synthetic instantiation examples, sanitized parsing-pipeline source, and `verify_release.py`) are released under CC-BY 4.0. Body-text literals derived from 3GPP CR/TR/TS PDFs are redistributed with explicit 3GPP source attribution, following the academic redistribution practice established by [11,23].

Verifiable boundary. The `verify_release.py` script (Section 7) verifies the `real_world_mini` synthetic instantiation contains only synthetic identifiers (no real company-name leak from the operational KG into the synthetic example set).

G Process-KG Data Quality and SHACL Conformance

The cross-WG process KG (`examples/process_kg/`) is shipped as three union-compatible Turtle files: `ls_routing.ttl` (195K triples, 26,791 LSs), `cr_routing.ttl` (1.25M triples, 192,967 CRs), and `ran1_tdoc_metadata.ttl`

(991K triples, 123,678 TDocs). Together they form a 2.44M-triple union graph that conforms to `shapes/spectra-core.shacl.ttl` with zero violations under pyshacl. Note: the schema graph (`ontology/spectra.ttl`, 887 triples) and this 2.44M-triple process KG measure different things—the 887 number counts *schema axioms* (class/property declarations), while 2.44M counts *instance triples* populated against that schema.

Three population-time conventions and one upstream-data observation, derived directly from inspection of the released TTLs and the source RAN1–RAN5 KGs, govern how the schema's lifecycle classes are instantiated; readers querying the release should be aware of them:

- **LS `originatedFrom` for outgoing LSs is implicit.** In 3GPP practice, an outgoing Liaison Statement (`direction = "out"`) has the host Working Group as its source by definition — the originating WG is the one tabling the LS. The Phase-1 RAN loaders therefore record `originatedFrom` only for incoming LSs (`direction = "in"`; coverage 51.9–96.6% across RAN1–RAN5, depending on the availability of an explicit *Original LS* reference on the cover sheet). For outgoing LSs the host WG is recoverable from the LS's hosting KG.
- **LS `sentTo` reflects upstream coverage.** 1,556 of 26,791 LSs (5.8%) carry no `sentTo` edge in the released union, mirroring an empty *To:* cell on the source TDoc cover sheet. SHACL therefore does not assert `sh:minCount` on `LSShape.sentTo`; consumers requiring a complete recipient list should treat absence as upstream-metadata gap rather than schema violation.
- **Tdoc `presentedAt` can be multi-valued under re-tabling.** 207 of 123,678 RAN1 TDoc instances (0.17%; identical pattern observed RAN-wide) appear in the cover-sheet records of more than one Meeting because the same TDoc was re-tabled at a subsequent meeting under the same identifier. `TdocShape.presentedAt` therefore omits `sh:maxCount`; queries treating `presentedAt` as functional should add an `ORDER BY` or aggregate clause.
- **Source companies are kept verbatim throughout.** Both the process-KG (`examples/process_kg/`) and the SpectraCQ benchmark (`cqs/spectra_cq_v1.0/`) retain real company names matching the public 3GPP per-meeting `TDOC_List.xlsx` and TDoc cover pages. No anonymization is applied because the names are already public.

The corresponding SHACL shape relaxations (`LSShape.originatedFrom`, `LSShape.sentTo`, `CRShape.modifies`, `TdocShape.presentedAt`: `sh:maxCount` removed) are all annotated inline in `shapes/spectra-core.shacl.ttl` with the rationales above.

CR-pack TSG plenary linkage. CR-pack approval at the TSG plenary is currently captured by the literal `tsgMeeting` attribute on `CRPack` (`xsd:string`, e.g., "RAN#103"); a structural `approvedAt` OP linking `CRPack` to a `TSGPlenary` $\sqsubseteq$ `Meeting` subclass is staged for v1.1 to support *approved/postponed/rejected* CR-pack analytics directly in the schema.